



COURSE INFORMATION SHEET

Session: 2026

Course Title: Distributed Database Systems

Course Code: CS 606

Credit Hours: 2+1

Semester: VI

Pre-Requisite: Database Management Systems

Instructor Name: Umair Ahmed

Email and Contact Information: umair.nazir21@gmail.com

Office Contact Hours:

- **Monday 11:00 am to 01:00 pm**

Mode of Teaching:

- Interactive lectures with problem-solving sessions
- Board and Pseudocode-Based Teaching
- Case-Based and Application-Driven Learning
- Formative Assessment and Feedback
- Hands-on Programming in lab.
- Project based on learning in the class.



COURSE DESCRIPTION

This course provides a systematic study of fundamental principles and architectures essential for building high-availability, scalable distributed database systems. It focuses on the design, implementation, and management of distributed data stores, including sharding, replication strategies, distributed query processing, and hashing techniques. The course also emphasizes key distributed concepts such as consensus algorithms, consistency models, transaction isolation, fault tolerance, and the trade-offs outlined by the CAP theorem. Through theoretical understanding and hands-on laboratory work, students develop the ability to evaluate distributed storage technologies (SQL, NoSQL, and NewSQL), analyze system performance, and architect resilient solutions for modern data-intensive applications.

COURSE OBJECTIVE:

This course aims to develop a strong understanding of fundamental data structures and their practical applications in solving computational problems. Students will learn to design, implement, and analyze data structures such as arrays, linked lists, stacks, queues, trees, graphs, and hash tables, with an emphasis on efficiency and algorithmic performance. The course also focuses on applying appropriate data structures to real-world problems, enhancing problem-solving skills, and preparing students for advanced courses in algorithms, software development, and system design.

COURSE OUTLINE:

● Module 1: Foundations & Distributed Architecture

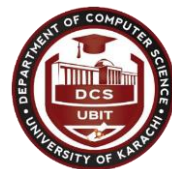
- Introduction to Relational & Non-Relational Databases: Review of ACID properties, indexing, execution plans, and the transition from centralized to distributed systems.
- Distributed Database Concepts & Architectures: Shared-everything vs. shared-nothing vs. shared-disk models; homogenous vs. heterogeneous databases.
- Data Partitioning & Sharding: Horizontal and vertical fragmentation, range partitioning, hash partitioning, and data allocation strategies.

● Module 2: Data Distribution, Indexing & Time

- Replication Strategies: Single-leader (active-passive), multi-leader, and leaderless replication; synchronous vs. asynchronous replication trade-offs.



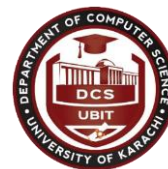
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- Distributed Hashing & Partitioning: Consistent hashing, ring topologies, virtual nodes, and dynamic cluster rebalancing.
- Distributed Indexing: Local vs. global indexes, partition-aware indexing, and search routing techniques.
- **Module 3: Distributed Transactions & Consistency**
 - Distributed Transactions & Atomic Commit: Two-Phase Commit (2PC), Three-Phase Commit (3PC), and the Saga pattern for long-running transactions.
 - Distributed Recovery & Logging: Write-Ahead Logging (WAL) in multi-node environments and distributed crash recovery (ARIES protocols).
 - Consistency Models & Distributed Isolation: Strict, sequential, causal, and eventual consistency; Multi-Version Concurrency Control (MVCC); snapshot isolation across nodes.
 - Consensus Algorithms: Leader election and state machine replication using Raft and Paxos protocols.
- **Module 4: Fault Tolerance, Storage Paradigms & Optimization**
 - Fault Tolerance & System Bounds: The CAP Theorem, PACELC Theorem, network partitions, split-brain scenarios, and failure detection.
 - Distributed Query Processing & Optimization: Query decomposition, localization, distributed joins (semi-join, bloom join), and global cost-based query optimization.
 - Modern Distributed Storage Paradigms: Comparing Distributed SQL (CockroachDB, Spanner), NoSQL (Key-Value, Document, Columnar), and NewSQL engines for modern data-intensive applications.



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COURSE LEARNING OUTCOMES (CLOs) and its mapping with GRADUATING ATTRIBUTES (GAs):

Course Learning Outcomes (CLOs):

CLO No.	Course Learning Outcomes (CLOs)	Bloom's Taxonomy	GAs Mapping
1	Explain the fundamental architectures, concurrency control mechanisms, consensus protocols, and consistency models used in distributed database systems.	C2	GA2: Knowledge for Solving Computing Problems
2	Apply distributed database strategies—such as sharding, replication, consistent hashing, and two-phase commit—to solve data distribution, availability, and scalability challenges.	C3	GA3: Problem Analysis
3	Analyze the trade-offs in distributed storage systems regarding consistency, availability, fault tolerance, and query performance under the constraints of the CAP and PACELC theorems.	C4	GA3: Problem Analysis

GRADING SCHEME

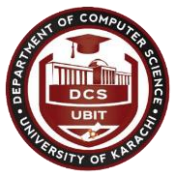
Assessment Tool	Weight
Mid Term	30%
Course Project / Lab	40%
Final Exam	30%

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tool	Weight	CLO1	CLO2	CLO3
Mid Term	30%	✓	✓	
Course Project / Lab	40%		✓	✓
Final Exam	30%	✓	✓	✓



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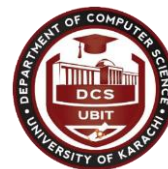


REFERENCE MATERIALS:

- Principles of Distributed Database Systems (4th Edition), Authors: M. Tamer Özsu and Patrick Valduriez, Role: Deep academic reference specifically focusing on distributed query optimization, 2PC/3PC commit protocols, and multi-site transaction management.
- Database System Concepts (7th Edition), Authors: Abraham Silberschatz, Henry F. Korth, and S. Sudarshan Publisher: McGraw-Hill, Role: Primary text for foundational concepts, indexing, relational algebra, and initial distributed database mechanics.



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LESSON PLAN

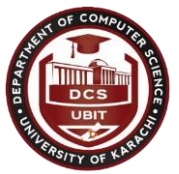
Course Title: Data Structures and Applications

Course Code: BSCS 451

Week	Topic	Details
1	Introduction to Database Foundations	Relational vs. non-relational model review; ACID properties; execution plans; indexing mechanics; transition from centralized to distributed DBs.
2	Distributed Architectures & Memory Models	Shared-everything, shared-disk, and shared-nothing architectures; homogenous vs. heterogeneous databases; multi-node memory allocation.
3	Data Partitioning & Sharding	Horizontal vs. vertical fragmentation; range-based vs. hash-based partitioning; data allocation and placement strategies.
4	Replication Strategies & Node Management	Single-leader (active-passive), multi-leader, and leaderless replication; synchronous vs. asynchronous replication trade-offs.
5	Distributed Hashing & Cluster Topology	Consistent hashing algorithms; ring topologies; virtual nodes; dynamic cluster rebalancing and node join/leave protocols.
6	Distributed Indexing & Search Routing	Local secondary indexes vs. global partition-aware indexes; scatter-gather search routing; bloom filters for key lookups.
7	Time & Event Ordering in Distributed Systems	Physical clock drift; logical clocks (Lamport Timestamps, Vector Clocks); Google TrueTime API and causal ordering.
8	Distributed Transactions & Atomic Commit (Part 1)	Multi-node transactions; Two-Phase Commit (2PC) protocol; coordinator failure, blocking scenarios, and recovery logs.



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9	Distributed Transactions & Atomic Commit (Part 2)	Three-Phase Commit (3PC) non-blocking protocols; Saga pattern for long-running microservice transactions.
10	Distributed Recovery & Write-Ahead Logging	Write-Ahead Logging (WAL) in distributed nodes; checkpointing; distributed ARIES crash recovery protocols.
11	Consistency Models & Distributed Concurrency	Strict, sequential, causal, and eventual consistency; Multi-Version Concurrency Control (MVCC) across nodes; snapshot isolation.
12	Consensus Algorithms	State machine replication; Paxos protocol; Raft consensus algorithm (leader election, log replication, safety).
13	System Bounds & Fault Tolerance	The CAP Theorem (Brewer's); PACELC Theorem; network partitions; split-brain scenarios; heartbeats and failure detectors.
14	Distributed Query Processing & Optimization	Query decomposition; distributed joins (Semi-join, Bloom join, shuffle joins); global cost-based query optimization.
15	Course Project Submission & Demonstration	Project Deliverable & Code Submission: Live demonstrations, benchmarking evaluations, architectural trade-off defenses, and peer evaluations.
16	Modern Distributed Storage Engines & Synthesis	Comparative analysis: Distributed SQL (Google Spanner, CockroachDB), NoSQL, and NewSQL engines; synthesis of course concepts.



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